1. INTRODUCTION

1.1 ORGANISATION BACKGROUND

Rio Grande Securities Ltd (RGS) is a medium-sized cybersecurity consultancy that offers security advisory services to small and medium-sized enterprises around the world. Historically the focus of RGS was on securing the network perimeter, but with the emergence of IIoT and IT/OT network convergence, the market has seen a huge demand for AI-driven security solutions for the Critical Infrastructure (CI) sectors such as energy, water, and transport.

1.2 RESEARCH PROBLEM STATEMENT

Signature-based IDS can only detect threats that have been catalogued before and cannot detect new OT attacks. This gap is empirically confirmed: 68.2% of surveyed practitioners rate current IDS effectiveness at 1 or 2 (mean=2.32/5), and 59.1% report IT/OT convergence incidents in the last three years (Stallings, 2023; Chen and Guestrin, 2016). RGS clients need technically superior, XAI-enabled, NIS2-compliant, and adversarial attack resilient alternatives based on AI.

1.3 RESEARCH AIM AND OBJECTIVES

Aim: To evaluate the feasibility, performance, and governance requirements of AI-powered IDS for CIP, commissioned by RGS.

- O1: Literature review of AI-driven IDS approaches and OT/CI gap identification.
- O2: Primary survey data (n=22) on AI adoption, XAI, and FL from CI professionals.
- O3: Algorithm benchmarking on SWaT and CIC-IDS datasets vs signature-based IDS baselines.
- O4: Concept drift impact analysis over 30-week ICS simulation.
- O5: Federated Learning evaluation as privacy-preserving cross-sector mechanism.
- O6: Evidence-based recommendations for RGS and CI-sector clients.

1.4 RESEARCH QUESTIONS

- RQ1: How does AI anomaly detection compare to signature-based IDS on novel OT threats?
- RQ2: What is the impact of XAI on operator trust and AI-IDS adoption in OT environments?
- RQ3: What are the principal barriers to FL adoption for cross-sector threat intelligence sharing?
- RQ4: How does concept drift affect long-term ML model performance in ICS environments?
- RQ5: What adversarial attack vectors pose the greatest risk to AI-based CI security systems?

1.5 SCOPE

This research targets CI sectors that face special security challenges due to the IT/OT convergence. It covers algorithm benchmarking (SWaT/CIC-IDS), a primary survey (n=22), and secondary literature review. Consumer IoT, quantum cryptography and live CI system testing are not included.